MSE 311 Thermodynamics of materials-Homework 10

Name: *Student ID:*

1. The activity coefficient of Zn in liquid Zn– Cd alloys at 435 °C can be represented as

$$\ln \gamma_{Zn} = 0.875 X_{Cd}^{2} - 0.30 X_{Cd}^{3}$$

Derive the corresponding expression for the dependence of ln γ_{Cd} on composition and calculate the activity of cadmium in the alloy of X_{Cd} = 0.5 at 43 5°C.

2. Considering the activity coefficients of the components of a binary solution could be represented by power series of the form

$$\ln \gamma_A = \alpha_1 X_B + \frac{1}{2}\alpha_2 X_B^2 + \frac{1}{3}\alpha_3 X_B^3 + \frac{1}{4}\alpha_4 X_B^4 + \cdots$$

$$\ln \gamma_B = \beta_1 X_A + \frac{1}{2}\beta_2 X_A^2 + \frac{1}{3}\beta_3 X_A^3 + \frac{1}{4}\beta_4 X_A^4 + \cdots$$

Are α_3 and β_3 equal to zero? What about α_4 and β_4? Why?

3. Calculate the heat required to form a liquid solution at 1356 K, starting with 1 mole of Cu and 1 mole of Ag at 298 K. At 1356 K, the molar heat of mixing of liquid Cu and liquid Ag is given by $\Delta H^M = -20{,}590\ X_{\mathrm{Cu}}X_{\mathrm{Ag}}$.

Hint: The total heat includes the heat required to increase temperatures of each components and the heat required due to the mixing of two components.

$$c_{p,Cu(s)} = 23.0 + \frac{1\times10^5}{T^2} \qquad [\text{J/mol/K}]$$

$$c_{p,Cu(l)} = 24.0 \qquad [\text{J/mol/K}]$$

$$c_{p,Ag(s)} = 21.0 + 9\times10^{-3}T + \frac{2\times10^5}{T^2} \qquad [\text{J/mol/K}]$$

$$c_{p,Ag(l)} = 30.0 \qquad [\text{J/mol/K}]$$

$$\Delta H_{melt,Cu,1356K} = 14000 \qquad [\text{J/mol}]$$

$$\Delta H_{melt,Ag,1234K} = 12000 \qquad [\text{J/mol}]$$

4. Melts in the system Pb–Sn exhibit regular solution behavior. At 473°C, $a_{\mathrm{Pb}} = 0.055$ in a liquid solution of $X_{\mathrm{Pb}} = 0.1$. Calculate the value of interaction parameter Ω for the system and calculate the activity of Sn in the liquid solution of $X_{\mathrm{Sn}} = 0.5$ at 500°C.